

8.3 UNIT TAX & SUBSIDY

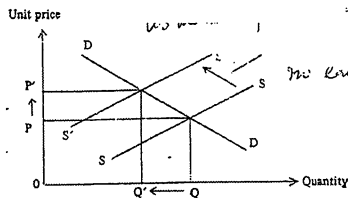
Microeconomics by Topic
8. Market Intervention

MARKING SCHEME

1990/CE/II/51 A	2003/CE/II/06 C (19%)	2010/CE/II/11 C (69%)	1995/CE/II/11 C	2010/CE/II/09 C (42%)
1991/CE/II/58 A	2003/CE/II/11 B (46%)	2010/CE/II/12 A (53%)	2002/CE/II/11 B (58%)	SP/DSE/I/20 D
1992/CE/II/25 A	2003/CE/II/12 D (25%)	2012/DSE/I/16 B (66%)	2004/CE/II/10 A (28%)	2014/DSE/I/16 C (48%)
1993/CE/II/29 B	2003/CE/II/42 A (32%)	2012/DSE/I/17 A (60%)	1990/CE/II/24 C	2015/DSE/I/10 D (32%)
1994/CE/II/08 B	2005/CE/II/09 B (68%)	2013/DSE/I/13 B (63%)	1990/CE/II/10 B	2015/DSE/I/12 B (48%)
1994/CE/II/09 C	2005/CE/II/11 B (47%)	2014/DSE/I/22 B (72%)	1996/CE/II/11 C	2016/DSE/I/19 A (43%)
1996/CE/II/07 A	2006/CE/II/11 B (39%)	2015/DSE/I/16 C (48%)	1999/CE/II/04 D	1994/CE/II/11 B
1998/CE/II/41 C	2006/CE/II/12 C (56%)	2017/DSE/I/16 B (64%)	2000/CE/II/08 B	2020/DSE/I/15 D
1999/CE/II/08 B	2008/CE/II/07 C (63%)	2018/DSE/I/22 A (74%)	2000/CE/II/11 D	2021/DSE/I/15 D
1999/CE/II/09 B	2008/CE/II/08 C (49%)	1991/CE/II/26 D	2001/CE/II/07 A	2021/DSE/I/16 B
2001/CE/II/08 C	2009/CE/II/13 B (47%)	1992/CE/II/26 B	2004/CE/II/11 B (34%)	2022/DSE/I/15 B
2022/DSE/I/21 D	2024/DSE/I/16 D	2024/DSE/I/17 B	2025/DSE/I/20 B	

1991/CE/I/1(b)
(i) Gasoline

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.



Candidates should indicate on the graph:

- leftward shift of S
- $P \uparrow$
- $Q \downarrow$

(1)
(1)
(1)

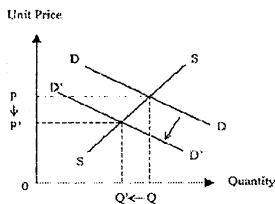
Elaboration:

The supply decreases (due to the gasoline tax).
Therefore, the equilibrium price increases, and
the equilibrium quantity decreases.

(1)
(1)
(1)

Microeconomics by Topic
8. Market Intervention

(ii) New cars



Candidates should indicate on the graph:

- leftward shift of D
- $P \downarrow$
- $Q \downarrow$

(1)
(1)
(1)

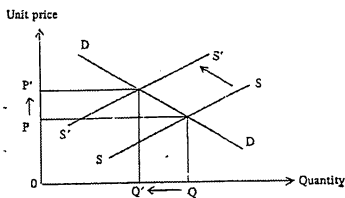
Elaboration:

The demand for new cars decreases, because new cars and gasoline are complements. Therefore, the equilibrium price decreases, and the equilibrium quantity of new cars decreases.

(1)
(1)
(1)
(1)

1992/CE/1/2(b)

(i) Hong-Kong made garments



Candidates should indicate on the graph:

- leftward shift of S
- $P \uparrow$ & $Q \downarrow$

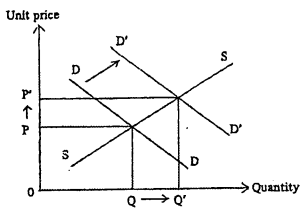
(1)
(1)

Elaboration:

The supply decreases (due to the import tax). Therefore, the equilibrium price increases, and the equilibrium quantity decreases.

(1)
(1)
(1)

(ii) US-made garments



Candidates should indicate on the graph:

- rightward shift of D
- $P \uparrow$ & $Q \uparrow$

(1)

(1)

Elaboration:

The demand increases (due to a rise in the price of substitutes).

(1)

Therefore, the equilibrium price increases, and

(1)

the equilibrium quantity increases.

(1)

1993/CE/1/4(c)

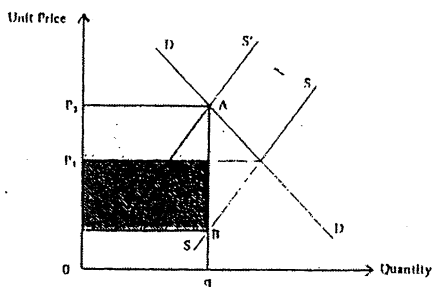


Diagram:

- AB = per unit tax
- correct shift of supply curve
- tax paid by consumers (white rectangle)
- tax paid by suppliers (black rectangle)

(1)

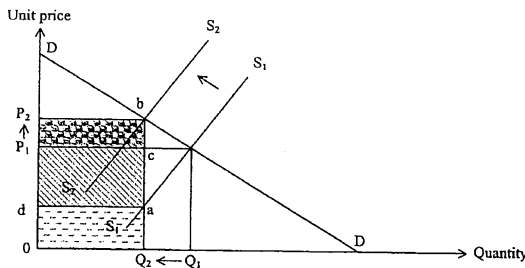
(2)

(2)

2004/CE/1/9(c)

(i) Condition: The elasticity of demand for cars is greater than the elasticity of supply.

(2)



Indicate in the diagram:

- supply curve shifts from S_1 to S_2
- sellers' tax burden = area acP_1d
- buyers' tax burden = area cbP_2P_1
- area $acP_1d >$ area cbP_2P_1

(1)

(1)

(2)

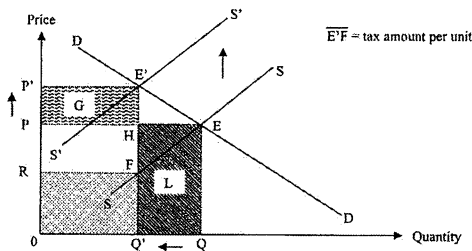
(ii) Indicate in the diagram:

- area $0Q_2ad$ is the net total sales revenue.

(1)

2007/CE/1/9(c)

(i)



Indicate in the diagram:

- upward shift of the supply curve
- a decrease in spending (= G) + increase in spending (= L)
- $G < L$

(1)

(1)

(1)

Verbal elaboration:

There is a decrease in supply.

(1)

The price increases and the quantity decreases.

(1)

The demand is elastic,

(1)

the percentage change in quantity demanded is larger than the percentage change in price.

(1)

(ii) Indicate in the diagram:

- sellers' revenue before tax (= area $0Q'EP$)
- the revenue net of tax (= area $0Q'FR$)

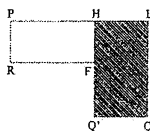
(1)

(1)

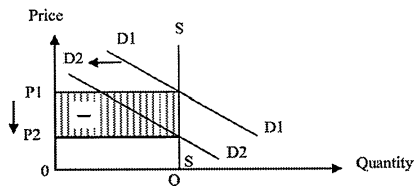
OR

- change in revenue = loss of area $Q'QEPRF$

(2)



2008/CE/1/9(b)(ii)



Indicate in the diagram:

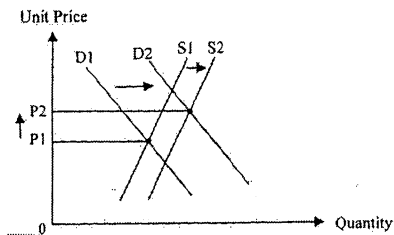
- leftward shift of the demand curve (1)
- market price falls from P_1 to P_2 (1)
- market value decreases from $P_1 \times Q$ to $P_2 \times Q$ (1)

Verbal elaboration:

The demand for stock market investments decreases,
due to lower expected net capital gain. (1)

There is a fall in the price of shares, and
the total market value $[P \times Q]$ falls. (1)

2009/CE/1/11(c)(i)



Indicate in the diagram:

- demand curve shifts to the right (1)
- supply curve shifts to the right (1)
- the new equilibrium price $>$ the original equilibrium price (1)

Verbal elaboration:

The demand for red wine increases (as it is more popular). (1)

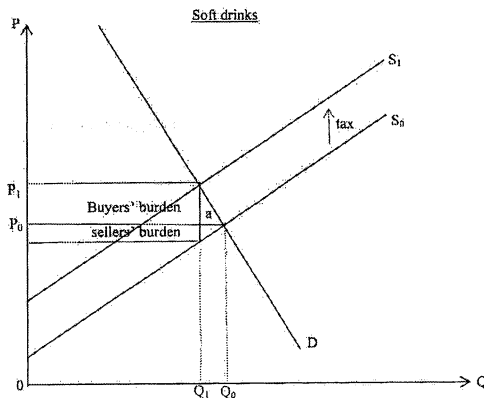
The supply of red wine increases (due to the tax cut). (1)

The increase in demand is larger than the increase in supply. (2)

2014/DSE/II/9

(a) The condition is that the elasticity of demand for the good is smaller than the elasticity of supply.

(1)



Indicate on the diagram:

- parallel upward shift of supply curve due to the per unit tax
- price increases and quantity drops
- correct positions of buyers' tax burden and sellers' tax burden
- buyers' burden > sellers' burden

(1)

(1)

(1)

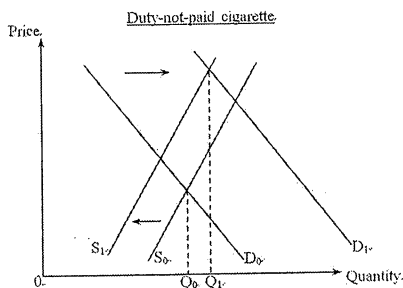
(1)

(b) Indicate on the diagram :

- reduction in consumer surplus: buyers' burden + area a

(1)

2015/DSE/II/11(d)



Indicate on the diagram:

- rightward shift of the demand curve
- leftward shift of the supply curve
- final equilibrium with a higher quantity transacted

(1)

(1)

(1)

Verbal elaboration:

On the one hand, due to the HK Customs' crack-down effort, the supply of duty-not-paid cigarettes would drop.

(1)

On the other hand, as duty-paid cigarettes and duty-not-paid cigarettes are substitutes, tobacco duty would increase the demand for duty-not-paid cigarettes.

(1)

It is thus possible that the total quantity of duty-not-paid cigarettes sold would increase,

(1)

if the increase in demand has a bigger effect on the quantity transacted than the decrease in supply.

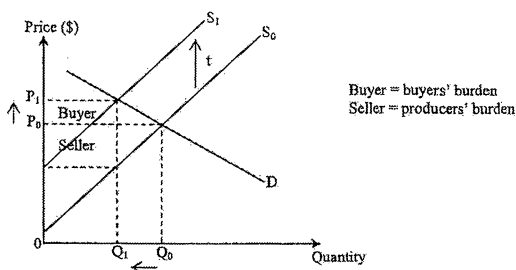
(1)

2016/DSE/II/10(c)

Condition: demand for solar panels is more elastic than its supply.

(2)

Chinese Solar panels sold in European Union



Indicate in the diagram:

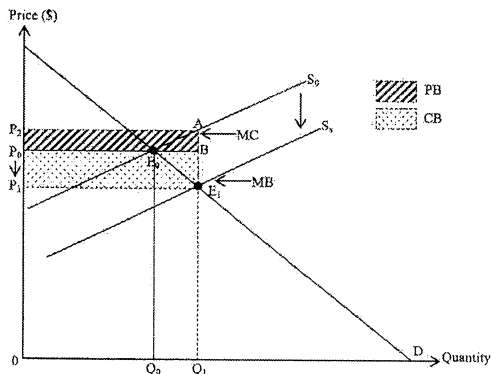
- upward shift of the supply curve
- P increases
- Q decreases
- correct position of buyers' burden and producers' burden
- buyers' burden < producers' burden

(1)
(1)
(1)
(1)
(1)

2017/DSE/II/11

(a) The condition is that the elasticity of demand for education services is lower than the elasticity of supply.

(2)



Illustrate in the diagram:

- parallel downward shift of supply curve due to the per unit subsidy
- correct position of new price (P_1)
- correct position of new quantity (Q_1)
- correct position of producer benefit and consumer benefit
- consumer benefit > producer benefit

(1)
(1)
(1)
(1)
(1)

(b) Illustrate in the diagram:

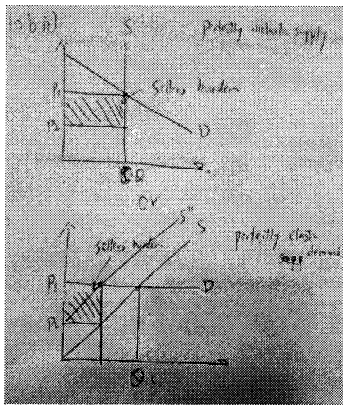
- correct position of marginal benefit
- correct position of marginal cost

(1)
(1)

Verbal elaboration:

At Q_1 marginal cost is larger than marginal benefit.

(1)



B) If the demand is perfectly elastic. Producer will bear all tax burden.

OR

If the supply is perfectly inelastic, Producer will bear all tax burden.

C) Consumer surplus after tariff = abc

Producer surplus before tariff = df_0

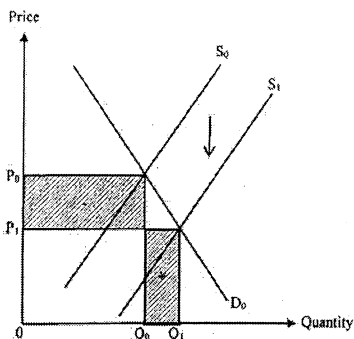
Producer surplus after tariff = hi0

(c) Verbal elaborations:

The provision of subsidy would result in an increase in the supply of masks. If demand is inelastic, the percentage decrease in price would be bigger than the percentage increase in quantity, so that total expenditure on masks would drop. (4)

Illustrate in the diagram:

- downward shift of supply curve (1)
- correct positions of the increase in total expenditure (+) and the decrease in total expenditure (-) (1)
- Area of (+) > Area of (-) (1)



- (d) Wearing mask protects oneself as well as lowering the chance of others being infected. Since the person wearing mask is not compensated by others (whose chances of being infected are reduced), there exists external benefit, with marginal social benefit exceeding marginal social cost. This implies that the number of people wearing mask is below the efficient level i.e. under-consumption of masks. Subsidy can increase the number of people wearing mask towards the efficient level (by lowering the marginal private cost), and thus narrow the gap between marginal social benefit and marginal social cost. (4)